Guard cell size and pore aperture influence stomatal closure kinetics

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Summary

- In fluctuating environments, the kinetics of stomatal opening and closing influence the balance between carbon gain and water loss. Smaller guard cells may respond faster to fluctuating environmental conditions because of their greater surface area for osmolyte flux relative to cell volume. A related hypothesis is that operational stomatal conductance (g_{op}) is often well below its theoretical maximum (g_{max}) because at this stomatal aperture, guard cell volume is poised to change rapidly with small changes in turgor pressure.
- We analyzed 2,124 estimates of stomatal closure kinetics in response to an abrupt increase in vapor pressure deficit (VPD) among 29 diverse wild tomato populations in the genus *Solanum*.
- Leaves with small guard cells and a lower g_{op} to g_{max} ratio (f_{gmax}) closed faster, but explained variation in kinetic parameters at different levels of biological organization. Guard cell size had high phylogenetic heritability and varied relatively little within populations, whereas f_{gmax} varied mostly among individuals and between light intensity treatments.
- Smaller stomata can be speedier, but only if stomata are held at an aperture where they are responsive to changing turgor pressure. Selection on stomatal speed may influence not only anatomical traits like guard cell size, but also physiological controls on g_{op}.

Introduction

In nature, change is the only constant. One way that vascular plants (tracheophytes) cope with change is by adjusting stomatal pore aperture to optimize the trade-off between carbon gain and water loss (Cowan & Farquhar, 1977; Raven, 2002; Hetherington & Woodward, 2003). In the absence of physical limits and energetic tradeoffs, natural selection would favor plants that could instantaneously adjust stomatal conductance (g_{sw}) to perfectly track dynamic environmental conditions (Lawson *et al.*, 2010). In reality, stomatal responses take time, creating a lag between actual and optimal stomatal aperture. For example, many species use hydroactive feedback mechanisms to precisely control stomatal responses to changing vapor pressure deficit (VPD), the driving force for evaporation (Buckley, 2019), but hydroactive control takes time. Leaves must sense a change in VPD through as-yet-unknown mechanisms and then transduce a signal that causes efflux of osmotic solutes to alter guard cell turgor more than the surrounding epidermal cells. Abscisic acid (ABA) dependent responses require several minutes to accumulate or synthesize hormone pools *de novo* (Xie *et al.*, 2006; Bauer *et al.*, 2013; McAdam *et al.*, 2016; Desai & Stroock, 2025); ABA-independent stomatal responses require changes in gene expression and protein synthesis that take time (Jalakas *et al.*, 2021). Once anion channels are activated, the rate of change in stomatal aperture depends on the rate of change in guard cell turgor (dP_{g}/dt) and the sensitivity of aperture to change in turgor ($da_{\mathrm{S}}/dP_{\mathrm{g}}$). As we discuss below, these two factors (dP_{g}/dt and $da_{\mathrm{S}}/dP_{\mathrm{g}}$) may be influenced by guard cell size and the ratio of operational to maximum stomatal conductance (f_{gmax}), respectively.

Over a century ago Darwin (1898) observed that after an initial increase, leaf transpiration decreases in response to decreasing humidity. He correctly attributed changes in transpiration to changes in stomatal aperture and subsequent studies have confirmed that stomatal aperture responds to humidity (Lange *et al.*, 1971) and, more specifically, transpiration rate (Mott & Parkhurst, 1991; Monteith, 1995), which is the product of VPD and g_{sw}. The VPD is related to relative humidity (RH) but also temperature. In angiosperms and some pteridophytes, a sudden increase in VPD, usually induced by lower RH at a constant temperature, causes a transient increase in g_{sw}, as Darwin observed (Cowan, 1972; Buckley *et al.*, 2011; Westbrook & McAdam, 2021). After a period of several minutes, stomata close until the leaf reaches a new steady-state g_{sw}. These phases of stomatal response to VPD are often referred to as the "wrong-way response" (WWR) and "right-way response" (RWR), respectively. The normative 'wrong' and 'right' appellations refer to the presumed adaptive significance. At steady state, g_{sw} is negatively correlated with VPD, all else being equal, which is interpreted as an adaptive response to the marginal carbon cost of water (Givnish & Vermeij, 1976; Cowan & Farquhar, 1977; Buckley *et al.*, 2017b). Models built around this assumption generally predict steady state g_{sw} well, although the functional form and presumed fitness costs are areas of active research (Damour *et al.*, 2010; Medlyn *et al.*, 2011; Prentice *et al.*, 2014; Sperry *et al.*, 2017; Potkay *et al.*, 2025).

Stomatal responses are as much about the journey (kinetics) as they are about the destination (steady state). If stomatal response kinetics are too slow relative to the temporal grain size of environmental variability, then it would be futile for g_{sw} to track environmental variation since the leaf will constantly lag the environment, perpetually fighting the last war. This may explain why g_{sw} can be insensitive to short sun/shade flecks (Knapp & Smith, 1990) and shaded leaves are likely primed to cope with the hydraulic demands of sudden increases in light, temperature, and VPD (Schymanski *et al.*, 2013). If stomatal responses can track environmental variation, then minimizing the lag time can greatly improve resource-use efficiency (Lawson *et al.*, 2010). For example, optogenetic control of stomatal aperture in response to fluctuating light increases water-use efficiency in *Arabidopsis* compared to wild-type responses (Papanatsiou *et al.*, 2019). Therefore the rate of stomatal opening and closing is predicted to have a major effect on how natural selection optimizes stomatal control in variable environments (Buckley *et al.*, 2023).

Much of the research on stomatal kinetics focuses on light responses because light intensity is usually the most important, rapidly fluctuating environmental variable affecting photosynthesis. In nature, fluctuating light intensity simulataneously alters leaf temperature and VPD (e.g. Ishida *et al.*, 1992). We focus on stomatal closure kinetics in response to an abrupt increase in VPD, rather than light, for four reasons. First of all, the primary aims and *a priori* hypotheses of this study were not to study stomatal kinetic parameters and are described in Muir *et al.* (2025). The idea to opportunistically use g_{sw}-response curves generated during this study to test hypotheses about which traits influence stomatal kinetics came after the data had been collected, but not yet analyzed for this purpose. Second, VPD responses exhibit a similar functional form as light-responses, suggesting similar underlying mechanisms (Buckley, 2019). Factors that affect stomatal response kinetics to VPD likely generalize to other environmental variables. Third, humidity responses tend to dominate over and are faster than other environmental responses (Aasamaa & Sõber, 2011a,b), but there is variation among species (Merilo *et al.*, 2014). Thus, factors that constrain maximum rates of stomatal closure are most likely to be observed in response to VPD. Finally, the ecological salience of VPD responses is increasing as climate change warms the atmosphere (Grossiord *et al.*, 2020).

Fully describing stomatal responses to the environment is complex (Buckley *et al.*, 2003; Jezek *et al.*, 2021; Desai & Stroock, 2025), but the dynamics of g_{sw} in response to an abrupt environmental change is well described by relatively simple phenomenological functions (Vico *et al.*, 2011; McAusland *et al.*, 2016). A four-parameter equation describes stomatal response kinetics to a step change in light intensity across dozens of plant species (Woning *et al.*, 2026):

$$g_{\mathrm{sw,t}} = g_{\mathrm{f}} + (g_{\mathrm{i}} - g_{\mathrm{f}})e^{-\left(\frac{t}{\tau}\right)^{\lambda}}. \tag{1}$$

In this equation, $g_{\mathrm{sw,t}}$ is stomatal conductance at time t, g_{i} is the initial g_{sw} before the step change, g_{f} is the final g_{sw} after the step change, τ is a time constant that describes how quickly stomata respond to the step change, and λ is a shape parameter that describes how the response rate changes over time. When $\lambda = 1$, the response rate is constant over time and τ represents the time required for g_{sw} to reach 36.8% $(1/e)$ of the way between initial and final values. When $\lambda > 1$, the response rate increases over time, and when $\lambda < 1$, the response rate decreases over time. Woning *et al.* (2026) refer to λ as a lag-time parameter because empirically $\lambda > 1$. We refer to stomatal kinetic parameters τ (time-constant) and λ (lag-time) henceforth. The first goal of this study is to evaluate whether kinetics of the RWR after a step change in VPD are adequately described by Equation 1. If so, it suggests that some fundamentally similar mechanisms are shared between light and hydroactive stomatal responses (Grantz & Zeiger, 1986).

The main goal of this study is to test hypotheses about which stomatal anatomical traits affect kinetic parameters τ and λ. We consider two interconnected hypotheses that have thus far been treated in isolation. The first hypothesis is that smaller guard cells open and close faster because of their intrinsically greater surface area to volume ratio (Franks & Farquhar, 2007; Drake *et al.*, 2013; Lawson & Vialet-Chabrand, 2019; Woning *et al.*, 2026). For approximately cylindrical cell geometries, like that found in guard cells, surface area increases linearly with radius, whereas volume increases in proportion to the radius squared. Consequently, larger guard cells will require more time, all else being equal, to achieve a given change in turgor pressure and, hence, stomatal pore aperture. A plant with large guard cells can partially compensate for this geometric constraint by increasing the activity and/or density of ion and solute transporters in the guard cell plasma membrane (Homann & Thiel, 2002; Lawson & Blatt, 2014; Raven, 2014). However, compensation may be limited by the available membrane area and/or energetic costs of maintaining high transporter densities (Vico *et al.*, 2011). Empirical support for the hypothesis that smaller guard cells are faster is mixed and has primarily come from step changes in light (Drake *et al.*, 2013; Elliott-Kingston *et al.*, 2016; McAusland *et al.*, 2016; Kardiman & Ræbild, 2018; Haworth *et al.*, 2023; Brench *et al.*, 2026; Tang *et al.*, 2026). Meta-analysis of dozens of species suggests that stomatal size alone is weakly related to kinetics and modulated by guard cell shape (Woning *et al.*, 2026). The relationship between guard cell size and stomatal response kinetics to leaf water potential (Aasamaa *et al.*, 2001) and VPD (Kübarsepp *et al.*, 2020; Qie *et al.*, 2024) is less well studied, but similarly equivocal.

One factor that might complicate the relationship between size and speed is aperture. Stomatal aperture can vary from near 0 when guard cell turgor pressure is low and asymptotically approach g_{max} as guard cell turgor pressure approaches ∞ (Figure 1D). The relationship between guard cell turgor pressure and stomatal aperture is nonlinear (Franks & Farquhar, 2001) as cell walls become less elastic at greater turgor (Franks *et al.*, 2001). The nonlinear relationship between guard cell turgor and pore aperture implies that if the rate of turgor change is constant ($\frac{\mathrm{d}P}{\mathrm{d}t} = C$), the rate of change in aperture, and hence stomatal conductance, will vary depending on the initial aperture (a_0) relative to the maximum aperture (a_{max}). Hence, $\frac{\mathrm{d}a}{\mathrm{d}t} = f(a_0/a_{\mathrm{max}})$. When initial aperture is high, g_{sw} will respond more slowly than when initial aperture is low. Since we generally do not observe individual stomatal aperture, relating this idea to leaf-level g_{sw} requires scaling by stomatal density, biophysical constants, and morphological constants (e.g. Sack & Buckley, 2016). We will use the term "fraction of anatomical maximum stomatal conductance", symbolized as f_{gmax} (this is denoted as g_{ratio} by Xie *et al.* (2022)). This value should be proportional to the average aperture divided by its maximum aperture for any arbitrary stomatal density. Leaves operating at f_{gmax} closer to unity will, all else being equal, be slower to respond than leaves operating with a f_{gmax} close to zero, independent of stomatal density. Franks *et al.* (2012) hypothesized that selection on stomatal density would maintain typical operational stomatal conductance (g_{op}) relative to g_{max} at a sufficiently low value that g_{sw} would be sensitive to relatively small changes in guard cell turgor pressure. Consistent with this hypothesis, the g_{op}:g_{max} ratio is often near 0.25 (McElwain *et al.*, 2016; Murray *et al.*, 2020), well below $f_{\mathrm{gmax}} = 1$, and in a range where g_{sw} would be responsive to small changes in guard cell turgor pressure. There is likely variation within and among species and growth forms in the g_{op}:g_{max} ratio (Xie *et al.*, 2022; Ochoa *et al.*, 2024).

Putting these two hypotheses together, we predict that both guard cell size and f_{gmax} will influence stomatal kinetics, but possibly at different scales of biological organization. Guard cell size tends to vary less than stomatal density or aperture at many biological scales. For example, on a mature leaf, guard cell length is nearly constant even as cell volume and pore aperture change with turgor (Willmer & Fricker, 1996; Franks *et al.*, 2001). Compared to stomatal density, guard cell size can be less variable (Lammertsma *et al.*, 2011; Bucher *et al.*, 2016) and evolve more slowly between species (Muir *et al.*, 2023). If there is relatively little plastic or genetic variation among individuals within a species, then most of the variance in stomatal kinetics within species cannot be explained by variation in guard cell size. In contrast, f_{gmax} can vary immensely within and among individuals because stomatal aperture responds dynamically over the day and in response to environmental variation. Because guard cell size and aperture typically vary at different levels of biological organization, we predict that guard cell size likely explains more variation in stomatal kinetics among species, whereas f_{gmax} will explain more variation within species. Within species variation can arise from either genetic differences among individuals or environmental plasticity. We will also test if adaxial stomata respond faster than abaxial stomata (Haworth *et al.*, 2018; Wall *et al.*, 2022; Woning *et al.*, 2026) and, if so, whether this difference can be explained by variation in guard cell size and/or f_{gmax} between surfaces.

One challenge in determining whether traits covary at higher levels of organization is that statistical procedures to account for phylogenetic nonindependence can obscure ecologically relevant covariation caused by natural selection (Westoby *et al.*, 1995; Uyeda *et al.*, 2018). For example, in a meta-analysis of stomatal kinetic responses to light, the explanatory power of traits like guard cell size and stomatal ratio

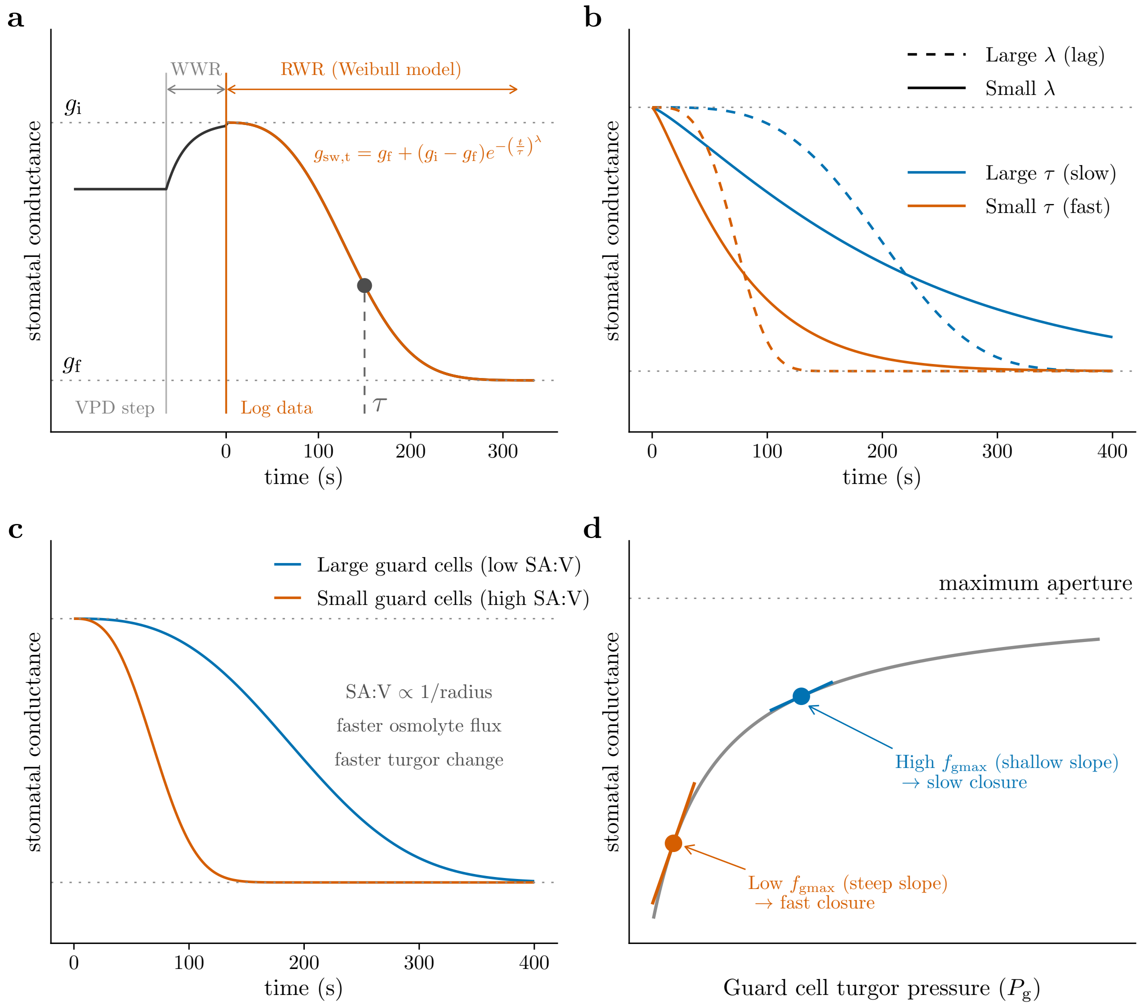

Figure 1: **Conceptual overview of the two hypotheses linking stomatal anatomy to closure kinetics.** (A) Stomatal conductance (g_{sw}) in response to a step increase in vapor pressure deficit (VPD) typically exhibits a transient wrong-way response (WWR), in which g_{sw} briefly rises, followed by the right-way response (RWR), in which g_{sw} declines to a lower steady state. The RWR is well described by a Weibull function (dashed line; Equation 1) with initial conductance g_i, final conductance g_f, and kinetic parameters τ and λ. (B) The time constant τ sets the overall speed of closure, and the lag-time λ determines whether the response rate is roughly constant ($\lambda \approx 1$; solid) or accelerates over time ($\lambda > 1$; dashed). (C) Smaller guard cells have a greater surface area to volume ratio (SA:V; schematic cross-sections shown at right) and are predicted to exchange osmolytes more quickly across the plasma membrane, yielding a faster change in guard cell turgor pressure and faster closure. (D) The relationship between guard cell turgor pressure (P_g) and g_{sw} is nonlinear and saturating because guard cell walls become less elastic at higher turgor (Franks & Farquhar, 2001; Franks *et al.*, 2001). Consequently, the same change in turgor pressure produces a larger change in g_{sw} when stomata are at low f_{gmax} (blue; steep tangent slope) than at high f_{gmax} (orange; shallow tangent slope), predicting slower closure when f_{gmax} is high.

declined in phylogenetic compared to nonphylogenetic regression (Woning *et al.*, 2026). This finding may be caused in large part because grasses have small stomata, high adaxial:abaxial stomatal ratio, and rapid stomatal kinetics. Are these associations a coincidence caused by shared ancestry or ecologically important covariation that helps explain how grasses adapt to their environment? Recent advances in phylogenetic comparative methods use hierarchical, multiresponse phylogenetic models (Muir *et al.*, 2023; Westoby *et al.*, 2023; Halliwell *et al.*, 2025) to tease apart "conservative trait correlations" that act at higher levels of organization (e.g. among species) from trait correlations that explain variation at lower levels of organization (e.g. among individuals within species and plastic responses). We use this approach to test whether guard cell size and f_{gmax} explain variation in stomatal kinetics at different levels of biological organization. Throughout, we use guard cell length (l_{gc}) as a proxy for size (Sack & Buckley, 2016).

The four main questions we address in this study are:

1. Are VPD kinetics adequately characterized by the same functional form in Equation 1 as light responses?
2. Are l_{gc} and f_{gmax} positively associated with τ?
3. At what biological levels of organization do l_{gc} and f_{gmax} vary and influence τ?
4. Are abaxial-only stomatal responses slower than whole leaf stomatal responses?

As noted above, we are opportunistically analyzing data from an experiment designed to answer different questions. Before analyzing the kinetic data, we planned to test whether 1) guard cell size would influence τ and 2) that τ would be greater on the abaxial-only leaves based on a recent meta-analysis of stomatal responses to light (Woning *et al.*, 2026). We made the decision to test for an association between f_{gmax} and τ after exploratory analyses revealed a potential association. Adding hypotheses after some results have been analyzed, also known as "Hypothesizing After Results Known" (HARKing), can lead to reports of spurious associations that are not repeatable in later studies (Kerr, 1998; Smaldino & McElreath, 2016). Therefore, our conclusions about the effect of f_{gmax} on stomatal kinetics should be interpreted with caution until they are replicated.

Materials and Methods

Populations and phylogeny

We grew 29 populations of wild tomato (Table S1), including representatives of all described species of *Solanum* sect. *Lycopersicon* and sect. *Lycopersicoides* (Peralta *et al.*, 2008), as described in Muir *et al.* (2025). These populations were selected because they encompass the breadth of climatic variation in the wild tomato clade (Pease *et al.*, 2016). Due to constraints on growth space and time, we spread out measurements over 61.1 weeks. Replicates within population were evenly spread out over this period to prevent confounding of temporal variation in growth conditions with variation among populations. For phylogenetic comparative analyses, we used the RAxML whole-transcriptome concatenated phylogeny based on 2,745 100-kb genomic windows from Pease *et al.* (2016). Two of our populations were not in this tree. We used *S. galapagense* accession LA1044 in place of LA3909. *S. pennellii* accession LA0750 was added as sister to LA0716. The node separating LA0750 from LA0716 was placed half-way between the next deepest node.

Plant growth conditions and light treatments

A thorough description of plant growth conditions can be found in Muir *et al.* (2025), therefore we summarize the most important information here. Seeds provided by the Tomato Genetics Resource Center germinated on moist paper in plastic boxes after soaking for 30-60 minutes in a 50% (volume per volume) solution of household bleach and water, followed by a thorough rinse. We transferred seedlings to cell-pack flats containing Pro-Mix BX potting mix (Premier Tech, Rivière-du-Loup, Quebec, Canada) once cotyledons fully emerged, typically within 1-2 weeks of sowing. We grew seeds and seedlings for both sun and shade treatments under the same environmental conditions (12:12 h, 24.3:21.7 °C, 49.6:58.4 RH day:night cycle). LED light provided PPFD $= 267\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ (Fluence RAZRx, Austin, Texas, USA) during the germination and seedling stages.

Immediately prior to starting growth light intensity treatments (sun and shade), we transplanted seedlings to 3.78 L plastic pots containing 60% Pro-Mix BX potting mix, 20% coral sand (Pro-Pak, Honolulu, Hawaiʻi, USA), and 20% cinders (Niu Nursery, Honolulu, Hawaiʻi, USA) with slow release NPK fertilizer following manufacturer instructions (Osmocote Smart-Release Plant Food Flower & Vegetable, The Scotts Company, Marysville, Ohio, USA). Percentage composition is on a volume basis. We watered to field capacity three times per week to prevent drought stress. Seedlings were randomly assigned in alternating order within population to the sun or shade treatment during transplanting. The average daytime PPFD was $761\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ and $115\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ for sun and shade treatments, respectively.

Humidity response curves

We selected a fully expanded, unshaded leaf at least six leaves above the cotyledons during early vegetative growth. This typically meant that plants had grown in light treatments for ≈ 4 weeks, ensuring they had time to sense and respond developmentally to the light intensity of the treatment rather than the seedling conditions (Schoch *et al.*, 1980).

We measured stomatal conductance over time after a step change in humidity at a constant leaf temperature, which we refer to this as a humidity-response curve. We measured humidity-response curves using a portable infrared gas analyzer (LI-6800PF, LI-COR Biosciences, Lincoln, Nebraska, USA). During gas exchange measurements, light-acclimated plants were placed under LEDs dimmed to match their light treatment. We collected four humidity-response curves per leaf, an amphistomatous (untreated) curve and a pseudohypostomatous (treated) curve at high light-intensity (PPFD $= 2000\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$; 97.8:2.24 red:blue) and low light-intensity (PPFD $= 150\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$; 87.0:13.0 red:blue). We always measured high light-intensity curves first because photosynthetic downregulation is faster than upregulation in these species. Pseudohypostomatous leaves are the same as the amphistomatous leaves except that gas exchange through the upper (adaxial) surface is blocked by a neutral density plastic (propafilm). For brevity, we hereafter refer to these as "pseudohypo" and "amphi" leaf types. Comparing amphi and pseudohypo leaf types enabled us to test whether stomatal kinetics are different for ab- and adaxial stomata on the same leaf. To compensate for reduced transmission, we increased incident PPFD for pseudohypo leaves by a factor 1/0.91, the inverse of the measured transmissivity of the propafilm. We also set the stomatal conductance ratio, for purposes of calculating boundary layer conductance, to 0 for pseudohypo leaves following manufacturer directions. To control for order effects, we alternated between starting with amphi or pseudohypo measurements. We made measurements over two days. On the first day, we measured high and low light-intensity curves for either amphi or pseudohypo leaves; on the second day, we measured high and low light-intensity curves on the other leaf type.

In all humidity-response curves, we acclimated the focal leaf ambient CO_2 ($C_\text{a} = 415\,\mu\text{mol}\,\text{mol}^{-1}$), $T_\text{leaf} = 25.0\,°\text{C}$, high light (PPFD $= 2000\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$), and high relative humidity (RH $= 70\%$; VPD $\approx 0.82\,\text{kPa}$) until g_sw reached its maximum. After that, we scrubbed water vapor from the incoming air using silica desiccant as quickly as possible, resulting in a water vapor concentration in the incoming air to be on average $0.36\,\text{mmol}\,\text{mol}^{-1}$. This treatment induced rapid stomatal closure, but it did not standardize chamber air humidity because leaf transpiration introduced water vapor. Consequently, the RH was systematically higher at high light intensity, in sun plants, and amphi leaf types (Table S3) because the g_sw was typically greater in these conditions. In all treatment conditions, the final RH ($7.62\% - 20.4\%$) and VPD ($2.5\,\text{kPa} - 2.92\,\text{kPa}$) elicited rapid stomatal response and is comparable to treatments used in similar experiments (Merilo *et al.*, 2014; Kübarsepp *et al.*, 2020; Hsu *et al.*, 2021; Qie *et al.*, 2024), but was not intended to mimic natural environmental variability. We started logging data after the transient WWR and continued logging data until g_sw reached its nadir (Figure 1A). We then acclimated the leaf to low light (PPFD $= 150\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$) and RH $= 70\%$ before inducing stomatal closure with low RH and logging data again.

Prior to analysis, we removed unreliable and unusable data points, removed points not at instrument steady state, removed the hysteretic portion of each curves at low g_sw, removed outliers within each curve, removed replicates with no overlap between amphi and pseudohypo curves from the same leaf, and thinned redundant data points within each curve. The rationale and procedure for each cleaning step is described in Muir *et al.* (2025). Additionally, we excluded stomatal kinetic parameter estimates from four curves where τ was estimated to be greater than 1097 s, far exceeding the typical estimates (see Results). After these steps, the total number of humidity response curves analyzed is given in Table S2. The median number of curves per population per treatment was 9 and the median number of points per curve was 26. We extracted the posterior distribution for τ and λ from each fitted curve to calculate the parameter estimate (median) and uncertainty (standard deviation) for subsequent analyses.

Stomatal anatomical traits l_{gc} and f_{gmax}

We estimated the stomatal density and l_{gc} on ab- and adaxial surfaces from all leaves used for humidity response curves. We used standard methods described fully in Muir *et al.* (2025). For each surface we calculated the anatomical maximum stomatal conductance (g_{max}) at a reference leaf temperature of 25 °C following Sack & Buckley (2016) as:

$$g_{\mathrm{max}} = bmdl_{\mathrm{gc}}/\sqrt{2}.$$

The biophysical and morphological constants b and m are:

$$b = \frac{D_{\mathrm{wv}}}{v}, \text{ and}$$

$$m = \frac{\pi c^2}{j^{0.5}(4hj + \pi c)},$$

where D_{wv} is the diffusion coefficient of water vapor in air and v is by the kinematic viscosity of dry air. We assumed $D_{\mathrm{wv}} = 2.49 \times 10^{-5}$ m^2 s^{-1} and $v = 2.24 \times 10^{-2}$ m^3 mol^{-1} (Monteith & Unsworth, 2013). For kidney-shaped guard cells like those in wild tomatoes, $c = h = j = 0.5$. The $g_{\mathrm{max,ratio}}$ is the g_{max} of the adaxial surface divided by the sum total g_{max} of both surfaces. To estimate f_{gmax}, we divided the initial g_{sw} estimated from the parameter g_{i} in Equation 1 from each humidity response curve by the anatomical g_{max} for that leaf. The estimated g_{i} was very close to the maximum g_{sw} observed near the beginning of each curve (Figure S2), suggesting that the estimated g_{i} was a good proxy for stomatal conductance at the time of the step change in humidity. We used the anatomical g_{max} for the surface(s) measured in each curve to calculate f_{gmax}. For amphi curves, we used the sum of adaxial and abaxial g_{max}; for pseudohypo curves, we used only the abaxial g_{max}.

Estimating kinetic parameters

We estimated τ and λ for each response curve by fitting Equation 1 to a time course of g_{sw} values using Bayesian nonlinear regression. All models were fit using Bayesian HMC sampling in the probabilistic programming language *Stan* (Stan Development Team, 2026) using the *R* package **brms** version 2.23.0 (Bürkner, 2017). We used *CmdStan* version 2.38.0 and **cmdstanr** version 0.9.0 (Gabry *et al.*, 2025) to interface with *R* version 4.5.3 (R Core Team, 2026). We sampled 1000 post-warmup draws from the posterior distribution on a single chain. We adjusted the number of sampling iterations and thinning interval for each curve so that the Gelman-Rubin convergence statistic ($\hat{R}$) (Gelman & Rubin, 1992) was less than 1.05 and the bulk effective sample size (ESS) was greater than 400 for all model parameters, and there were fewer than 10 divergent transitions. To answer Question 1, we assessed model fit using Bayesian R^2 (Gelman *et al.*, 2019) implemented in **brms**.

Estimating effects of stomatal anatomy on kinetic parameters

We first used model selection to determine whether anatomy (l_{gc} and f_{gmax}) affected kinetic parameters (λ and τ; Question 2) and at what level of biological organization (Question 3). From the selected model, we used parameter estimates and 95% confidence intervals to test the predictions of our hypotheses. We interpreted nonzero fixed effects of l_{gc} and f_{gmax} on λ and τ as evidence that these anatomical traits influence variation in stomatal kinetics among individuals. We interpreted nonzero phylogenetic correlations among traits as evidence that these traits covary among populations. Finally, we used mediation analysis to test whether l_{gc} and/or f_{gmax} contribute to plasticity in stomatal kinetics. Table 1 summarizes the statistical criteria for determining whether and at what level of biological organization stomatal anatomical traits affected kinetic parameters. The subsections below describe specific procedures for deciding whether those are met.

Bayesian multiresponse phylogenetic models

Multiresponse models can disentangle whether traits are associated among populations, individuals, and/or treatments that induce plasticity. The response variables in our models are τ, λ, l_{gc}, and f_{gmax}. All models included a subset of explanatory variables that were integral to the experimental design or necessary to account for statistical nonindependence. As explanatory variables, we included fixed effects of manipulative treatments: growth light intensity (sun and shade); measurement light intensity (high and low); and leaf type (amphi and pseudohypo). All models also included population as both a phylogenetically structured random effect and a non-phylogenetic random effect following Halliwell *et al.* (2025). Prior to fitting, the phylogenetic (co)variance matrix was rescaled as a correlation matrix (De Villemereuil & Nakagawa, 2014). For traits measured multiple times within the same leaf (f_{gmax},

Table 1: Criteria used to determine whether and at what level of biological organization stomatal anatomical traits affected kinetic parameters.

Level of organization	Criteria supporting an association between stomatal anatomy and kinetics
Among individuals	• Anatomical parameters as fixed effects improve model fit (lower LOOIC) • Anatomical effects on kinetics are in the predicted direction • Confidence intervals of anatomical effects do not overlap zero
Plasticity among individuals	• Plasticity in anatomy mediates plasticity in stomatal kinetics
Among populations	• Confidence intervals of phylogenetic correlation among traits do not overlap zero • Phylogenetic covariance is in the predicted direction

λ, τ) we included a random effect of individual to account for repeated measures. We accounted for uncertainty in τ and λ using the standard deviation of the posterior distribution of each estimate (see 'Humidity response curves' for details). We modeled residual (co)variance, a combination of among-individual (co)variance and measurement error, as a multivariate Student t-distribution, which is more robust to extreme values than the multivariate normal distribution (Gelman *et al.*, 2014). To account for heteroskedasticity, bounded measurements, and to linearize relationships, l_{gc}, λ, and τ were log-transformed prior to analysis whereas f_{gmax} was logit-transformed. All models were fit with *Stan* following the same methods described above for fitting Equation 1 to the humidity response curves.

Phylogenetic correlation among traits

We inferred that traits covary among populations if the 95% confidence intervals of the phylogenetic correlation between traits did not overlap zero and the correlation was in the predicted direction. For example, a positive phylogenetic correlation between l_{gc} and τ would be consistent with the prediction that species with larger guard cells have slower stomatal kinetics. Phylogenetic covariance specifically indicates long-term evolutionary covariation among traits because our model simultaneously accounts for trait associations among individuals, shared plastic responses, and residual covariance. To test for possible causation, we used the inferred phylogenetic trait covariance matrix to estimate partial correlations between anatomical and kinetic variables (Halliwell *et al.*, 2025). We interpret a significant partial correlation between variables as a plausible causal hypothesis.

Model selection and parameter estimation of individual-level effects

To infer whether individual-level variation in f_{gmax} and l_{gc} influence kinetic parameters, we used the leave-one-out cross-validation information criterion (LOOIC) to compare the fit of models using the *R* package **loo** version 2.9.0 (Vehtari *et al.*, 2017). We compared models with all permutations of f_{gmax} and l_{gc} influencing λ and τ. We considered all models within two ΔLOOIC standard errors of the minimum LOOIC (Table 2) to be in the set of plausible models to generate posterior predictions for hypothesis testing. Within each plausible model, we assessed whether a fixed effect was significantly different from zero based on the 95% confidence intervals.

Mediation analysis to test for plastic effects

We used causal mediation analysis (Pearl, 2022) to test whether plasticity in τ was mediated by plasticity in f_{gmax} or l_{gc} in response to measurement light intensity, growth light intensity, and curve-type treatments. We inferred the direct effect of treatments on τ from model coefficients. We estimated the mediated effect of treatments using the product of the effect of treatment on mediator (f_{gmax} or l_{gc}) and the effect of the mediator on τ (Imai *et al.*, 2010). We determined significant direct and mediator effects based on whether the 95% confidence intervals of the effects were different from zero. We calculated effect sizes as the point estimate (median of the posterior) divided by the standard error (standard deviation of the posterior).

Variance decomposition and phylogenetic heritability

One explanation for why traits have effects at different levels of organization is that variance in those traits is concentrated at those different levels of organization. To evaluate this, we decomposed the variance in response variables (λ, τ, f_{gmax}, l_{gc}) into phylogenetic, population (nonphylogenetic), and individual levels. The phylogenetic variance component quantifies trait evolution between populations congruent with the phylogenetic relationships, whereas the population (nonphylogenetic) component quantifies variance among populations independent of phylogenetic relationships (Lynch, 1991; De Villemereuil & Nakagawa, 2014). We estimated these components from the associated random-effect standard deviations for each response variable. The among-individual variance components quantify variance among individuals within the same population and treatment. We estimated the among-individual component as the sum of the individual random-effect and residual standard deviations. It was not possible to estimate individual random-effect in l_{gc} with our study design because we only measured average guard cell length once per leaf. The among-individual component is biased upwards because it also includes measurement error, which cannot be separately estimated with our experimental design. The phylogenetic heritability is equivalent to the phylogenetic variance component and is estimated following Lynch (1991) and De Villemereuil & Nakagawa (2014) as:

$$h^2_{\text{phy}} = \frac{\sigma^2_{\text{phy}}}{\sigma^2_{\text{phy}} + \sigma^2_{\text{pop}} + \sigma^2_{\text{ind}}},$$

where σ^2_{phy}, σ^2_{pop}, σ^2_{ind} are the phylogenetic, population (nonphylogenetic), and among-individual variance components, respectively. Because phylogenetic heritability was estimated after accounting for fixed treatment effects, estimates obtained in controlled environments likely overestimate phylogenetic heritability in natural environments, where plasticity is a larger component of trait variation.

Effect of adaxial stomata on kinetics

If adaxial stomata respond faster than abaxial stomata (Question 4), we predicted that τ should be greater in pseudohypo leaves compared to amphi. We evaluated this prediction by estimating the fixed effect of leaf type (amphi vs pseudohypo) on τ in the multiresponse model. We did not estimate adaxial-only kinetics because it not part of the original experimental design (Muir *et al.*, 2025).

Results

The time-constant τ explains most variation in stomatal closure kinetics

Stomatal conductance (g_{sw}) decreased rapidly in response to a step change in humidity. The model in Equation 1 fit the data extremely well (Figure S1). The mean and minimum Bayesian R^2 across all curves was 0.9994 and 0.9802, respectively. Consider a typical leaf in the experiment characterized by the median time constant (τ) and lag time (λ) among wild tomato populations in all treatments. After humidity decreased and the transient WWR elapsed, it took on average 127 s for g_{sw} to decrease half-way (t_{50}) from its initial (g_{i}) to final (g_{f}) steady state value. In terms of the kinetic model parameters, most variation among leaves was due to difference in the time constant rather than the lag time. In the previous example, increasing λ from its median to its maximum estimated value among populations increased the t_{50} by only 7.2 s. In contrast, increasing τ from its median to maximum value increased t_{50} by 88 s. Hence, we primarily focus on treatments and anatomical parameters affecting τ.

Variation in guard cell length and aperture

Guard cell length (l_{gc}) differed among populations and varied to a lesser extent between growth light intensity treatments and leaf surfaces. In contrast, stomatal conductance as a fraction of maximum conductance (f_{gmax}) was most strongly determined by measurement light intensity. Excluding treatment effects, phylogenetically structured differences among populations (i.e., phylogenetic heritability) accounted for 63.3% [95% CI: 21.4 to 83.9%] of the variance in l_{gc} (Figure 2, Table S4). There was also developmental plasticity in l_{gc}. On average, l_{gc} increased by 12.1% [95% CI: 11.1 to 13.0%] in the sun treatment (Figure 3a; Table S7). The average l_{gc} of pseudohypo leaves was -5.8% [95% CI: -6.6 to -5.0%] shorter than amphi leaves (Figure 3a; Table S7). This was not because of plasticity, as we measured amphi and pseudohypo curves on the same leaf. Rather, this was a composition effect. Adaxial stomata are larger than abaxial stomata in these species (Figure S3), so the average l_{gc} in pseudohypo leaves was lower than in amphi leaves because the former only included abaxial stomata.

The phylogenetic heritability of f_{gmax} was low, accounting for 16.9% [95% CI: 1.7 to 40.1%] of the variance (Figure 2, Table S4). High measurement light intensity increased f_{gmax} by an average of 0.85 logit units [95% CI: 0.82 to 0.87] (Figure 3b; Table S7). For example, if $f_{gmax} = 0.1$ in low light, the typical f_{gmax} for the same leaf acclimated to high light was 0.21. Sun leaves had on average -0.28 logit units [95% CI: -0.35 to -0.21] lower f_{gmax} than shade-grown plants (Figure 3b; Table S7). The pseudohypo leaf type had on average 0.21 logit units [95% CI: 0.18 to 0.24] higher f_{gmax} than untreated, amphistomatous leaves (Figure 3b; Table S7). This indicates that stomatal aperture was more open in leaves where gas exchange through adaxial stomata was blocked.

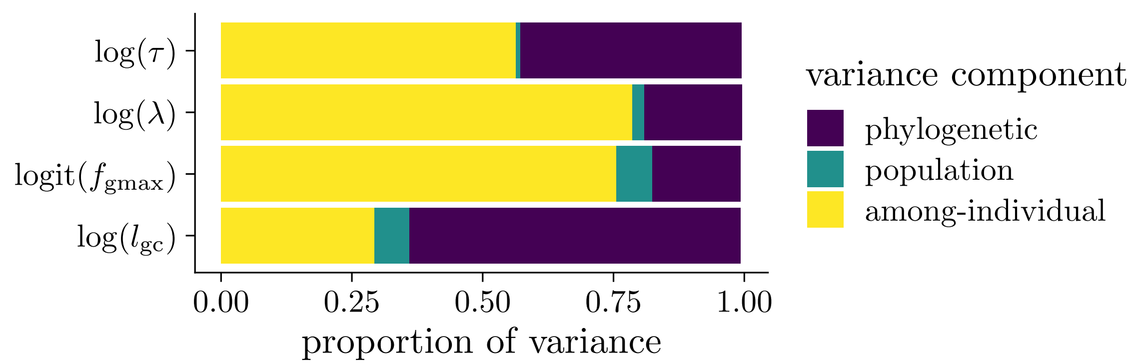

Figure 2: **Traits vary at different levels of biological organization**. A preponderance of the variation in guard cell length (l_{gc}) and the time-constant τ is phylogenetically structured among populations. In contrast, most of the variance in stomatal conductance as a fraction of anatomical maximum stomatal conductance (f_{gmax}) and the lag-time (λ) is among individuals. Nonphylogenetic variation among populations was low for all traits.

Variation in stomatal closure kinetic parameters

Estimates of stomatal closure kinetic parameters for each curve are in Table S5 and estimates for each population are in Table S6. The time-constant τ exhibited moderate phylogenetic heritability, accounting for 42.2% [95% CI: 18.6 to 61.2%] of the variance, whereas λ varied more within populations (Figure 2, Table S4). Some light treatments also affected kinetic parameters. In sun plants, τ was on average

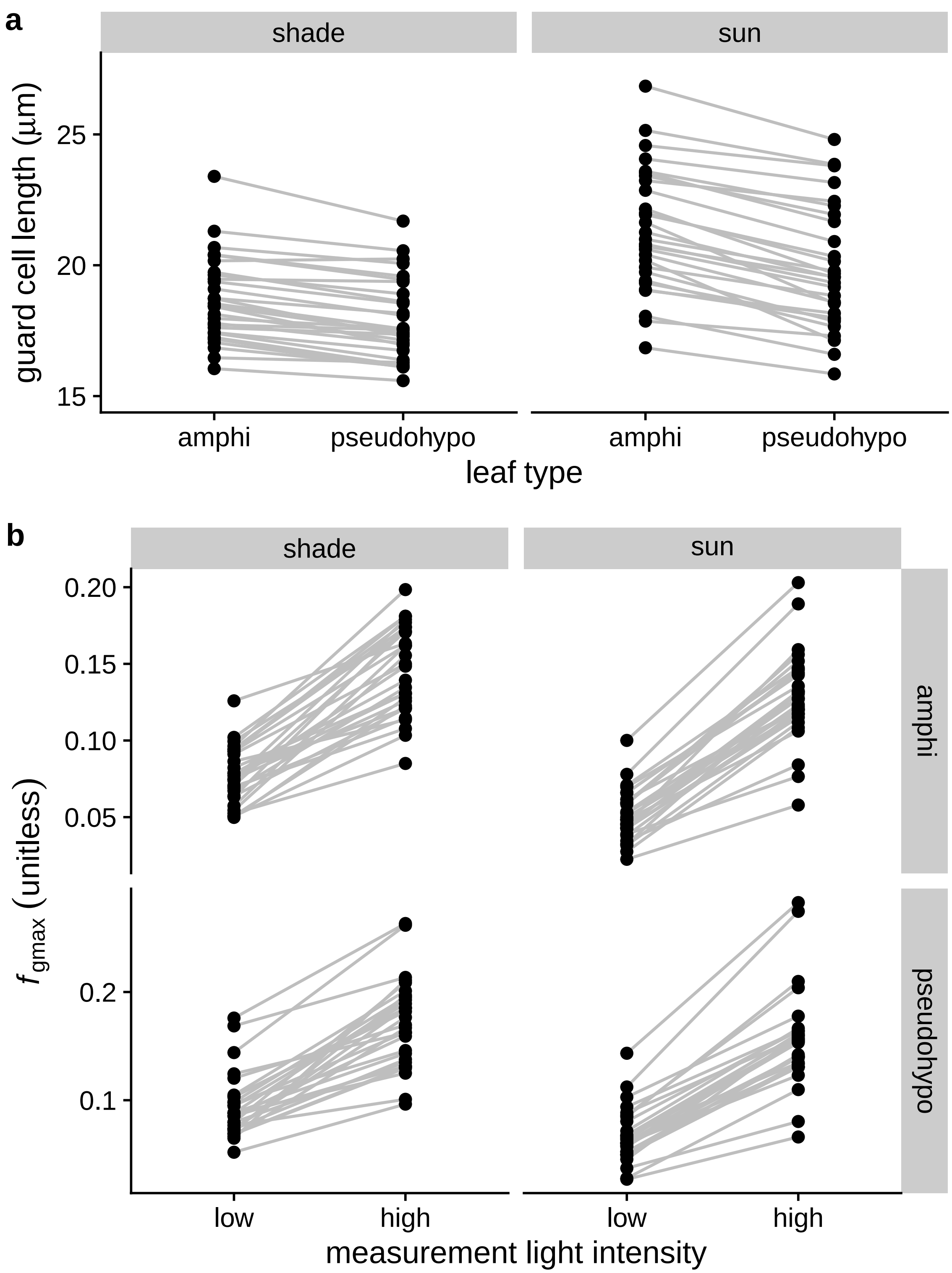

Figure 3: Caption on following page.

Figure 3: (previous page) **Guard cell length (l_{gc}) and stomatal conductance as a fraction of maximum conductance (f_{gmax}) vary among populations and treatments.** In both panels, black points are mean trait value per population within a particular treatment. Gray lines connect values within populations across treatments. (a) Most of the variation in l_{gc} is among populations, but developmental plasticity to growth light intensity tends to increase l_{gc} in sun (right facet) compared to shade (left facet) grown plants. There is no plasticity to leaf type (amphi vs. pseudohypo), but the average l_{gc} is slightly lower in pseudohypo leaves because adaxial stomata tended to be larger than abaxial stomata on the same leaf. (b) Measurement light intensity has the most direct effect on f_{gmax} because stomatal aperture increases under high light intensity. To a lesser extent, f_{gmax} varies among populations, is slightly lower in sun-grown plants, and slightly higher in pseudohypo leaf types.

5.56% [95% CI: 1.05 to 10.4%] higher than in shade plants after accounting for other explanatory variables (Figure 4, Table S7). Within the same leaf, increasing the measurement light intensity from 150 µmol m^{-2} s^{-1} to 2000 µmol m^{-2} s^{-1} significantly increased τ by 17.9% [95% CI: 11.6 to 24.8%] (Figure 4, Table S7). The time lag λ responded to growth, but not measurement light intensity after accounting for other explanatory variables (Figure 4, Table S7). In sun plants, λ was on average 12.2% [95% CI: 10.1 to 14.2%] greater than in shade plants. High measurement light had little effect on λ, with an estimated increase of 1.17% [95% CI: -1.1 to 3.54%].

Guard cell length and f_{gmax} influence stomatal kinetics

Guard cell length and f_{gmax} affected stomatal closure kinetics in the directions predicted by our hypotheses, but they operated at different levels of biological organization. Among the models we tested, six were plausible in that their predictive accuracy was similar based on ΔLOOIC (Table 2). All models in the plausible set indicated among-individual level effects of f_{gmax} on τ and λ, as we discuss in more detail below. Models that included both individual and phylogenetic effects of l_{gc} had similar LOOIC values but we selected a model with only phylogenetic effects for two reasons (Table 2). First, in models that included both individual and phylogenetic effects, neither the phylogenetic correlation nor the fixed effect regression coefficient were significantly different from zero (results not shown). These parameters are highly colinear in the posterior (Figure S4) indicating difficulty in estimating both terms simultaneously and "smearing" out their confidence intervals (McElreath, 2020). Hence, the model with only phylogenetic effects is easier to interpret. The second reason we favor this model is that it is consistent with the high phylogenetic heritability in both l_{gc} and τ (Figure 2). Visually, we observed positive relationships among population mean trait values (Figure 5) but little relationship between l_{gc} and τ within populations (results not shown). When comparing many models simultaneously, differences

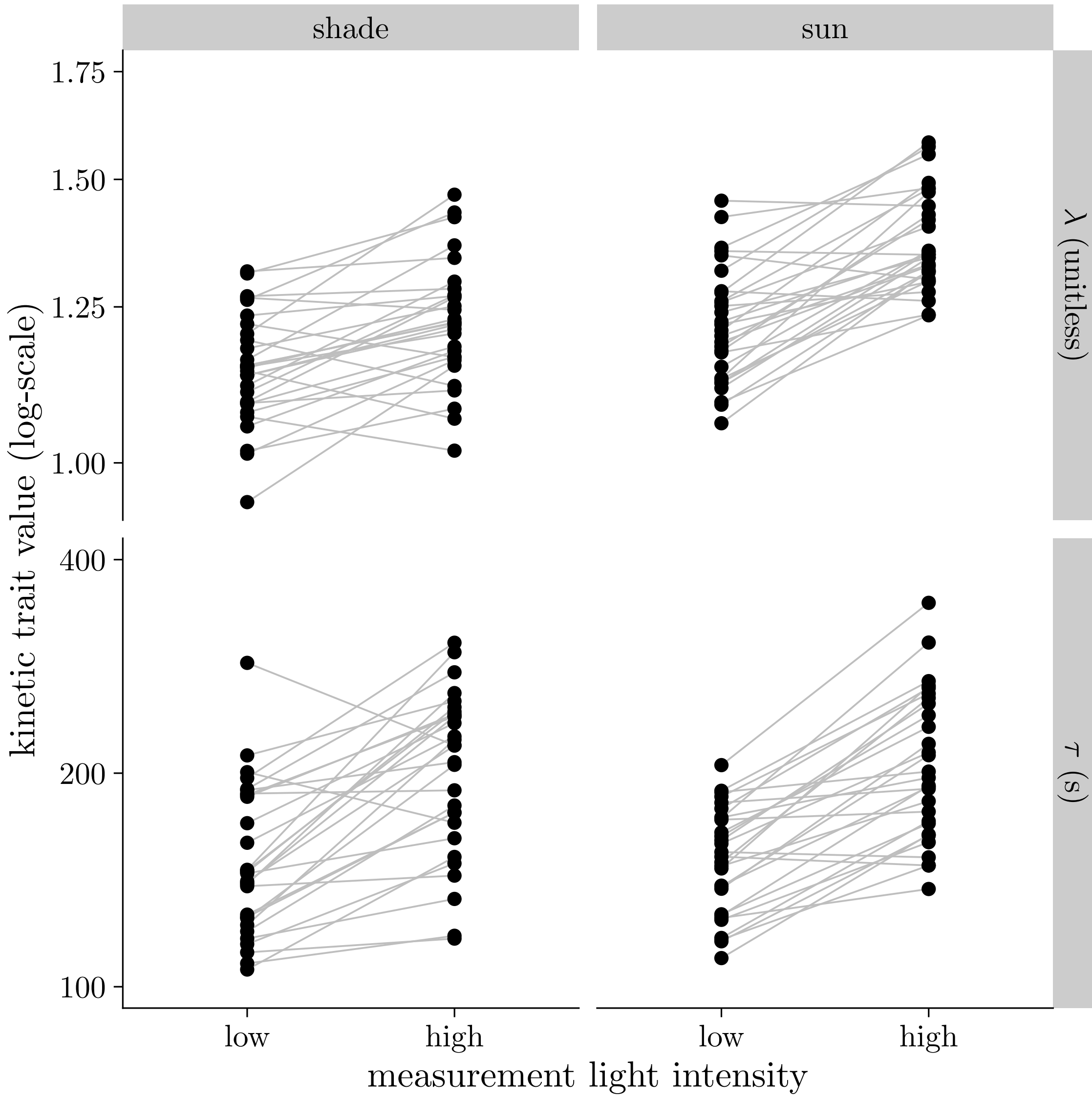

Figure 4: **Stomatal kinetics parameters λ (lag time; top facets) and τ (time constant; lower facets) vary among wild tomato populations.** Stomatal conductance decreases faster (lower τ) in response to a step change in vapor pressure deficit (VPD) in leaves when measured under low light intensity. The pattern was consistent for both shade (left facets) and sun (right facets) grown plants. The pattern for λ was qualitatively similar to that for τ. Each point is the average parameter value for one accession in that treatment combination. The growth and measurement light intensity treatments are described in the Materials and Methods section.

Table 2: Model comparison results. Models with a ΔLOOIC less than two standard errors (SE) from the minimum LOOIC are considered to have substantial support and are used for posterior predictions to test hypotheses about the effects of guard cell length (l_{gc}) and the stomatal conductance as a fraction of anatomical maximum stomatal conductance (f_{gmax}) on stomatal kinetic parameter λ (time lag) and τ (time constant). Unsupported models are in gray italic font. The model selected for subsequent inference is in bold. LOOIC stands for leave-one-out cross-validation information criterion.

		λ		τ	
ΔLOOIC	SE	f_{gmax}	l_{gc}	f_{gmax}	l_{gc}
0.00	0.00	✓	✓	✓	✓
1.38	**3.08**	✓		✓	
1.91	3.14	✓	✓	✓	
2.83	3.04	✓		✓	✓
16.16	10.83	✓	✓		
17.77	10.84	✓			
23.79	*10.76*	✓	✓		✓
24.29	*10.94*	✓			✓
46.64	*14.58*		✓	✓	
47.79	*14.48*		✓	✓	✓
49.54	*14.63*			✓	
50.34	*14.54*			✓	✓
68.81	*18.41*		✓		
71.66	*18.38*				✓
72.76	*18.48*				
75.18	*18.39*		✓		✓

in LOOIC performance can occur by chance (McLatchie & Vehtari, 2024), so judgement calls among models with similar performance can be necessary. The preponderance of evidence indicates that l_{gc} explains variation primarily among rather than within our populations. The results in the main text are based on estimates from the selected model.

Leaves with stomatal apertures closer to their maximum aperture closed more slowly than those less open (Figure 6). For example, increasing f_{gmax} from 20% to 30% resulted in a 10.7% [95% CI: 7.2 to 14.4%] increase in τ for a typical leaf in the reference treatment set (shade, low, amphi). The f_{gmax} also had a significant, positive, individual-level effect on λ (Table S7).

Populations that tended to have greater l_{gc} also tended to have greater τ as evidenced by a positive phylogenetic correlation (Table S8, Figure 5). However, we found little evidence that variation in l_{gc} explained τ or λ within populations. There were no other significant phylogenetic correlations

(Table S8). The only marginally significant partial correlation was between l_{gc} and τ ($\rho = 0.74$ [95% CI: -0.18 to 0.98]), indicating a plausible causal effect of l_{gc} on τ among populations.

Stomatal aperture mediates plasticity in τ

We imposed three treatments that influenced stomatal kinetics: growth light intensity, measurement light intensity, and leaf type. All three treatments directly affected τ and had indirect effects mediated by f_{gmax} (Figure 7). We did not estimate effects mediated by l_{gc} because the selected model did not retain direct effects of l_{gc} on τ (Table 2). The direct effect of the sun growth treatment increased τ by 5.58% [95% CI: 1.05 to 10.44%], but the indirect effect of lower f_{gmax} counteracted it by -5.06% [95% CI: -7.30 to -3.18%]. The direct effect of the pseudohypo treatment decreased τ slightly by -2.64% [95% CI: -4.98 to -0.15%], but the indirect effect mediated by greater f_{gmax} increased τ by 4.02% [95% CI: 2.64 to 5.49%]. In contrast, direct and indirect effects of high measurement light intensity increased τ. The direct effect increased τ by 17.90% [95% CI: 11.63 to 24.82%] and the indirect effect mediated by f_{gmax} increased τ by 17.31% [95% CI: 11.34 to 23.48%].

No evidence that adaxial stomata close faster than abaxial stomata

The time constant τ was slightly greater in pseudohypo leaves compared to the paired amphi leaf (results not shown), which would suggest that adaxial stomata close faster. However, pseudohypo leaves also had greater f_{gmax}, which slows stomatal closure, as discussed in the previous section. Once, the indirect effect of f_{gmax} on τ is accounted for, the direct effect of pseudohypo leaf type on τ disappears, indicating that the observed difference in τ between pseudohypo and amphi leaves is mediated by f_{gmax} (Figure 7). There is actually a tendency for τ to be slightly lower when only abaxial stomata are measured in pseudohypo leaves, as evidenced by the negative effect of this leaf type on τ (Table S7).

Discussion

The ability for stomata to adjust to fluctuating environmental conditions likely evolved in early land plants because of selection to optimize water use relative to carbon gain (Raven, 2002). Hydroactive control of stomatal aperture in angiosperms increases flexibility to modulate carbon gain and water loss depending on leaf water status (Buckley, 2019), perhaps enabling a competitive advantage in environments where light and evaporative demand fluctuate at high frequency (McAdam & Brodribb, 2012, 2015). We now understand the steady-state response of stomatal conductance (g_{sw}) to global increases in vapor pressure deficit (VPD) better but there is greater uncertainty surrounding kinetic responses to VPD (Buckley *et al.*, 2023). Here we considered two nonmutually exclusive mechanistic hypotheses for how stomatal anatomy influences stomatal closure kinetics. First, smaller stomata may

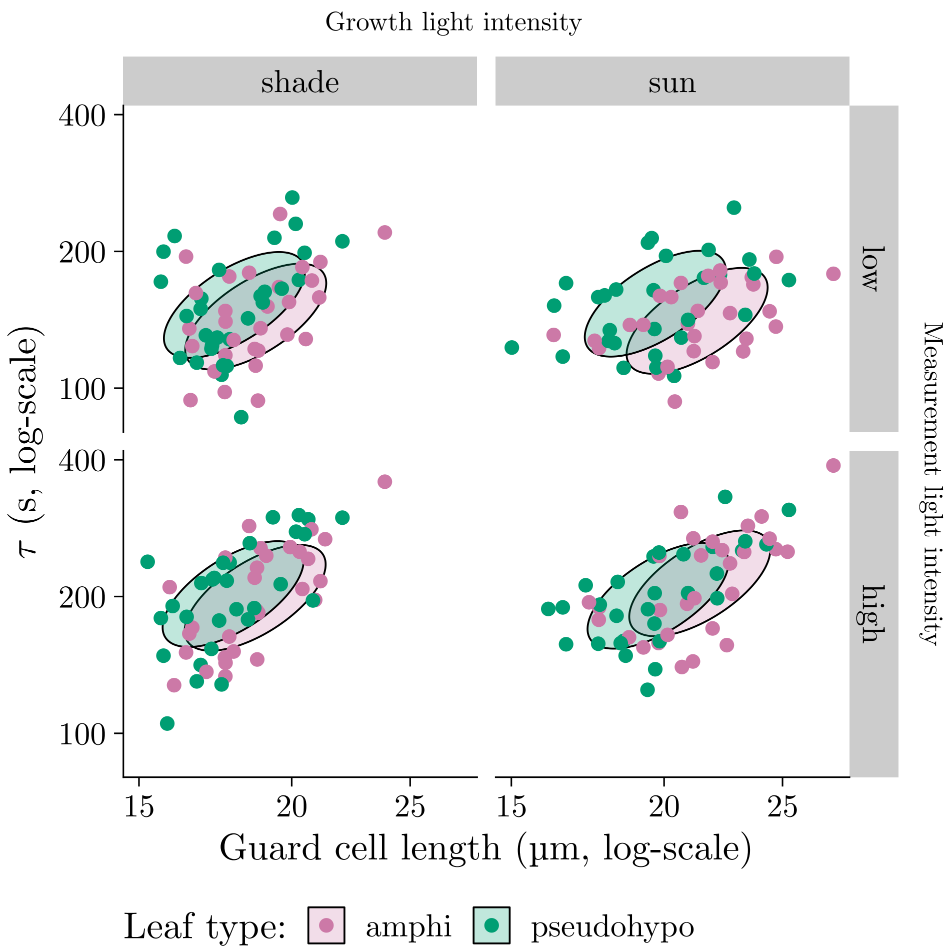

Figure 5: **Phylogenetic covariance between guard cell length (l_{gc}, x-axis) and the stomatal closure time-constant (τ, y-axis) indicates that populations with larger stomata tend to close more slowly in response to humidity.** The overall pattern was consistent across growth light intensity treatments (left and right facets), measurement light intensity treatments (top and bottom facets), and leaf types (point colors). The direction and magnitude of the ellipses are such that they should encompass 95% of the phylogenetically structured (co)variance in l_{gc} and τ on log-transformed scales. The location of ellipses are set to the median trait values in each treatment combination for visual comparison. Each point is the average parameter value for one population in that treatment combination. The growth and measurement light intensity treatments are described in the Materials and Methods section.

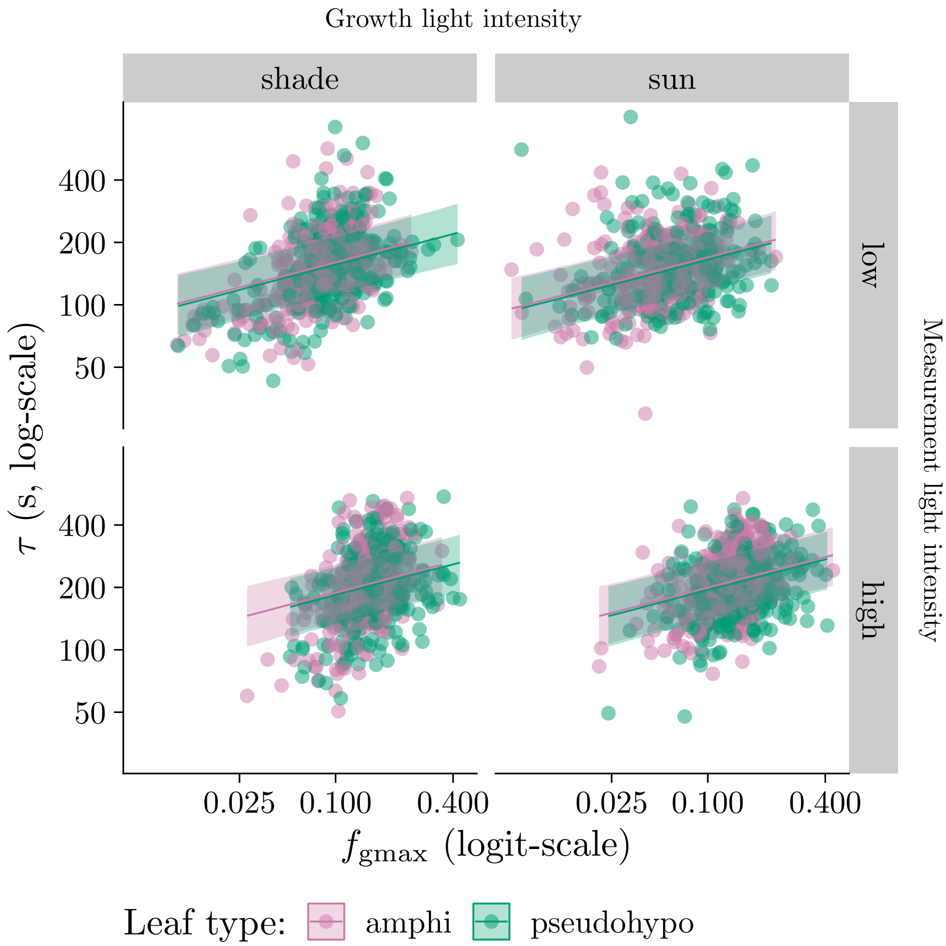

Figure 6: **Individual-level variation in stomatal apertures influences stomatal closures kinetics.** As the stomatal conductance as a fraction of anatomical maximum stomatal conductance (f_{gmax}, x-axis) increase, the stomatal closure time-constant (τ, y-axis) increases. The overall pattern was consistent across grow light intensity treatments (left and right facets), measurement light intensity treatments (top and bottom facets), and leaf types (point colors). Lines are estimated using linear regression along with 95% confidence ribbons. The growth and measurement light intensity treatments are described in the Materials and Methods section.

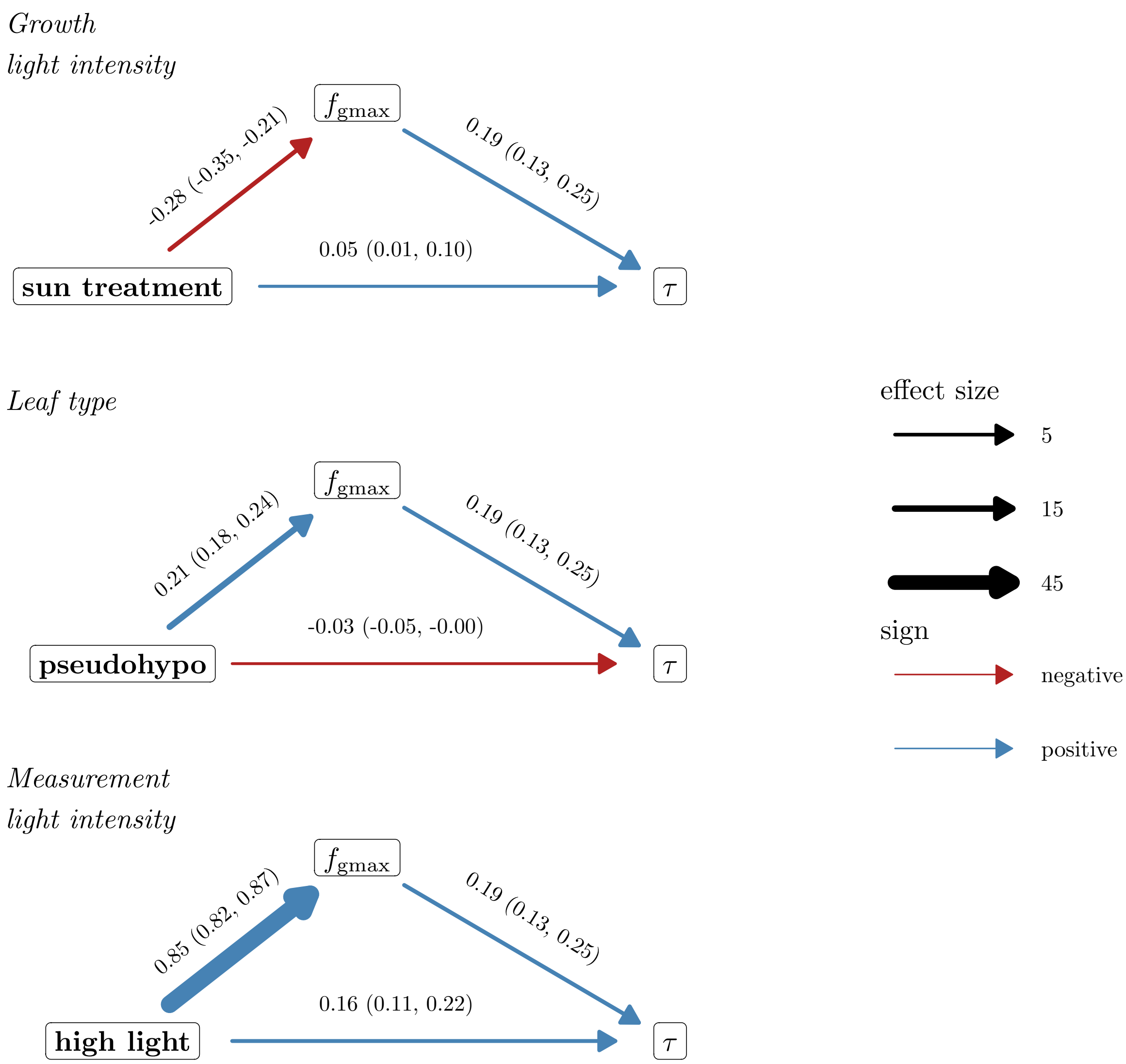

Figure 7: **Stomatal aperture as a fraction of anatomical maximum stomatal conductance (f_{gmax}) mediates plasticity in the time constant τ.** The three treatments are growth light intensity (top facet; shade vs. sun), leaf type (middle facet; amphi vs. pseudohypo), and measurement light intensity (bottom facet; low vs. high). Horizontal arrows at the bottom of each graph indicate the direct effect of the treatment level on τ; slanted arrows leading to or from f_{gmax} indicate mediated effects of the treatment on τ. Line thickness is proportional to the effect size and line color indicates whether the effects are negative (red) or positive (blue). Parameter estimates of the effect and 95% confidence intervals are above each line. The selected model did not retain direct effects of l_{gc} on τ and hence this variable could not be considered as a mediator.

close faster because of greater surface area to volume ratio (Drake *et al.*, 2013). Second, guard cells under higher turgor pressure respond more slowly because of reduced cell wall elasticity (Franks *et al.*, 2012). Based on these hypotheses, we predicted that leaves with smaller guard cells and operating further from their anatomical maximum g_{sw} (i.e., lower f_{gmax}) would have faster stomatal closure kinetics (lower τ). Analysis of 2,124 time courses of stomatal closure from 29 populations in multiple environments demonstrates that both guard cell size and stomatal aperture influence kinetics at different levels of biological organization. This new understanding suggests unexplored explanations for the relationships between stomatal size, density, speed, and operational g_{sw} (g_{op}) among angiosperms.

The kinetics of stomatal closure in response to VPD closely match response to light, suggesting similar underlying mechanisms. The lag-time model proposed by Woning *et al.* (2026) for light responses also describes humidity responses extremely well following the passive wrong-way response. The fitted models explained on average 99.94% of the variation in g_{sw} during closure (Figure S1). The formal mathematical similarity of light and humidity response might indicate that certain factors, such as guard cell size, influence stomatal kinetics responses in general. Alternatively, disparate underlying processes may be described by the same macroscale formalism. For example, humidity responses may be controlled by ABA accumulation or loss (McAdam *et al.*, 2016), whereas C_i-independent red-light responses can be influenced by accumulation or loss of photosynthetic products (Matthews *et al.*, 2020). These disparate processes may both have similar kinetic properties, but would not necessarily imply that VPD and light responses are strongly correlated. Future research comparing the correlation between light, VPD, CO_2, and other stomatal kinetic responses can determine the extent to which generic or specific factors determine stomatal opening and closing rates (Creese *et al.*, 2014; Kübarsepp *et al.*, 2020).

The positive association between l_{gc} and τ (Figure 5) is consistent with the hypothesis that smaller cells respond faster because they have a greater surface area to volume ratio for osmolyte flux (Franks & Farquhar, 2007; Drake *et al.*, 2013; Lawson & Vialet-Chabrand, 2019; Woning *et al.*, 2026). If correct, this implies that the maximum rate of stomatal closure is limited by membrane surface area. We do not have the data to evaluate the hypothesis mechanistically, but this should be tested in the future. An alternative explanation is that the correlation between l_{gc} and τ is not causal. The fact that l_{gc} did not explain much of the variation in τ among individuals within populations suggests that l_{gc} may not directly affect τ. However, the high phylogenetic heritability and, hence, low variability in l_{gc} within populations may also explain why we were unable to detect a within-population effect of l_{gc} on τ.

A consequence of greater stomatal aperture is that the pore closes more slowly when the environment changes. Much of the variation in τ among individuals within populations was explained by variation in f_{gmax} (Figure 6). This factor was positively associated with τ among individuals within light treatments, but also mediated much of the plastic effects of these treatments on τ (Figure 7). The rate of stomatal opening in response to light is associated with g_{op} in a sample of rainforest trees (Kardiman & Ræbild, 2018), which may indicate this is a more general phenomenon. The association between f_{gmax} and τ could be explained by the hypothesis that guard cell wall elasticity declines at greater turgor pressure (Franks *et al.*, 2012), like a rubber band that is stretched under tension.

The impact of f_{gmax} was most pronounced when comparing measurement light intensity treatments. Leaves acclimated to high light intensity were on average 111% more open than the same leaves at low light intensity in the reference treatment (shade, amphi). As a consequence, stomata closed more slowly in leaves acclimated to high light intensity, with a 17.3% greater time-constant on average in the reference treatment, based on parameter estimates from the selected model (Table S7). We did not assess whether f_{gmax} also affects stomatal opening and this should be measured to further test this hypothesis. The implication is that natural selection should favor g_{op} values that not only optimize instantaneous carbon gain and water loss in the present environment, but also carbon gain and water loss integrated through time in a fluctuating environment. We predict that rapid, temporally-uncorrelated fluctuating environments will favor leaves with greater f_{gmax} to slow kinetic responses and prevent overreacting to transitory changes that are likely to revert; slower, temporally-correlated changes will favor lower f_{gmax} to enable close tracking of the environment.

The absence of a direct effect of leaf type on τ suggests that adaxial and abaxial stomata close at similar rates when operating at similar apertures. The association between species with more adaxial stomata and faster stomatal closure (Haworth *et al.*, 2018; Woning *et al.*, 2026) may not be caused by adaxial stomata responding faster *per se*. Instead, selection may favor faster closure in leaves with adaxial stomata to prevent desiccation, causing increased abaxial stomatal speed as a byproduct. When there is a sudden increase in evaporative demand, such as from a sunfleck, open adaxial stomata could cause photosynthetically active palisade mesophyll to desiccate (Buckley *et al.*, 2017a). If stomata do not close quickly, this will lower water potential and impair photosynthesis (Tezara *et al.*, 1999). Hypostomatous leaves are less impacted because the palisade are hydraulically buffered from changing VPD (Buckley *et al.*, 2017a). If traits, such as guard cell size, that affect adaxial stomata kinetics similarly affect abaxial stomata, this could explain why leaves with greater stomatal ratio also have faster kinetics despite independent environmental responses on each surface (Mott, 2007; Wall *et al.*, 2022).

Our findings also suggest a novel explanation for the well-documented negative relationship between stomatal size and density across land plants (recently reviewed in Liu *et al.* (2025)). Conventional explanations for this relationship focus on optimizing g_{op} and minimizing area allocated to stomata, but not on kinetics. Our results point to an additional, previously unrecognized constraint. Smaller stomata do appear to close faster, consistent with the surface area to volume hypothesis, but this kinetic advantage is contingent on operating at a moderate or low f_{gmax}. If a leaf reduced guard cell size while maintaining the same g_{op}, the reduced $g_{\max}$ would push f_{gmax} higher, partially offsetting the speed benefit of smaller cells. The only way to simultaneously reduce guard cell size, maintain g_{op}, and keep f_{gmax} from rising is to increase stomatal density—raising $g_{\max}$ and thereby limiting f_{gmax}. This logic predicts a coupled evolution of small guard cells and high stomatal density, a pattern observed broadly across vascular plants (Franks & Beerling, 2009; de Boer *et al.*, 2016; Haworth *et al.*, 2023; Liu *et al.*, 2025). Although physical constraints on pore geometry, such as increased resistance to diffusion at very small apertures (Leuning, 1983; Hodgson *et al.*, 2010), likely impose a lower limit on guard cell size, selection to prevent operating at high f_{gmax} (Franks *et al.*, 2012) may also explain why optimal guard cell size varies.

Future research is needed to determine how stomatal kinetic responses measured in controlled conditions impact performance in nature. Our experimental protocol induced stomatal closure by abruptly switching incoming air to near-zero humidity, a perturbation far more severe than the gradual or partial VPD fluctuations leaves experience in the field. This approach is well suited for estimating the maximum rate of stomatal closure and comparing it among individuals and treatments, but it need not follow that the same anatomical and physiological factors that govern τ in our experiment also govern kinetic variation under more naturalistic conditions, where closure VPD may fluctuate more gradually. We hypothesize that ecological conditions which favor close tracking of fluctuating environments will select for leaves with traits like smaller guard cell size and low f_{gmax} that enable rapid kinetics.

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Competing interests

None declared.

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Author contributions

Conceptualization: C.D.M.; Data acquisition: C.D.M., W.S.L.; Data analysis and visualization: C.D.M.; Writing – original draft: C.D.M.; Writing – review & editing: C.D.M., W.S.L.

Data availability

All data and code are available on GitHub (https://github.com/cdmuir/solanum-kinetics). Data will be archived on Dryad upon publication; code will be archived on Zenodo upon publication.

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Supporting Information

Supporting tables

Table S1: Accession information of *Solanum* populations used in this study. The species name, accession number, collection latitude, longitude, and elevation. TGRC: Tomato Genetics Resource Center; mas: meters above sea level.

Species	TGRC accession	Latitude	Longitude	Elevation (mas)
S. arcanum	LA2172	-6.008	-78.858	662
S. cheesmaniae	LA0429	-0.644	-90.329	800
S. cheesmaniae	LA3124	-0.804	-90.042	1
S. chilense	LA1782	-15.267	-74.633	1000
S. chilense	LA4117A	-22.907	-67.941	3540
S. chmielewskii	LA1028	-13.883	-73.017	3000
S. chmielewskii	LA1316	-13.400	-73.906	2920
S. corneliomulleri	LA0107	-13.117	-76.383	60
S. corneliomulleri	LA0444	-13.433	-76.133	100
S. galapagense	LA0436	-0.953	-90.978	40
S. galapagense	LA1044	-0.284	-90.548	0
S. habrochaites	LA0407	-2.181	-79.884	70
S. habrochaites	LA1777	-9.550	-77.700	3216
S. huaylasense	LA1358	-9.533	-77.967	750
S. huaylasense	LA1360	-9.546	-77.929	1490
S. huaylasense	LA1364	-10.133	-77.383	2920
S. lycopersicoides	LA2951	-19.317	-69.450	2200
S. lycopersicoides	LA4126	-19.287	-69.396	3120
S. neorickii	LA1322	-13.483	-72.442	2380
S. neorickii	LA2133	-3.400	-79.183	1980
S. pennellii	LA0716	-16.225	-73.617	50
S. pennellii	LA0750	-14.775	-75.034	550
S. pennellii	LA3778	-14.775	-75.034	616
S. peruvianum	LA2744	-18.550	-70.150	400
S. peruvianum	LA2964	-18.028	-70.835	75
S. pimpinellifolium	LA1269	-11.483	-77.075	400
S. pimpinellifolium	LA1589	-8.433	-78.817	30
S. pimpinellifolium	LA2933	-1.442	-80.562	375
S. sitiens	LA4116	-22.159	-68.782	2960

Table S2: Total number of humidity response curves analyzed in this study broken down by treatments.

	Measurement light intensity				
	Low		High		
Growth light intensity	amphi	pseudohypo	amphi	pseudohypo	Total
shade	275	275	270	266	1,086
sun	245	245	274	274	1,038
Total	520	520	544	540	2,124

Table S3: Relative humidity (RH) and leaf vapor pressure deficit (VPD) during stomatal kinetics measurements systematically differed because of differences in stomatal conductance. Stomatal conductance tended to be greatest in sun-grown plants, measured at high light intensity, and untreated (amphistomatous). The mean values across replicate curves in each treatment conditions are given; within each curve we calculated the median RH and VPD across time points before calculating the mean among curves.

Growth light intensity	Measurement light intensity	Leaf type	RH (%)	VPD (kPa)
shade	low	pseudohypo	7.62	2.92
shade	low	amphi	8.25	2.91
shade	high	pseudohypo	13.3	2.74
shade	high	amphi	14.7	2.69
sun	low	pseudohypo	9.63	2.85
sun	low	amphi	11.1	2.81
sun	high	pseudohypo	16.8	2.62
sun	high	amphi	20.4	2.50

Table S4: The relative contribution of phylogenetic, among population (nonphylogenetic), and among-individual varation differs among stomatal traits. The phylogenetic component is equivalent to the phylogenetic heritability. The population component is the non-phylogenetic variance among populations. The among-individual variance component is the variation among individuals within a population after accounting for treatment effect. Estimates and 95% confidence intervals (CI) are estimated as the median and quantile intervals of the posterior distribution. f_{gmax}: stomatal conductance as a fraction of anatomical maximum stomatal conductance; l_{gc}: guard cell length; λ: lag time; τ: time constant.

component	% variance	95% CI
$\log(l_{\text{gc}})$		
phylogenetic	63.3	[21.4, 83.9]
population	6.7	[0.4, 34.5]
among-individual	29.3	[14.9, 48.5]
$\text{logit}(f_{\text{gmax}})$		
phylogenetic	16.9	[1.7, 40.1]
population	7.0	[1.6, 18.7]
among-individual	75.5	[55.4, 86.8]
$\log(\lambda)$		
phylogenetic	18.7	[1.9, 39.2]
population	2.3	[0.0, 12.1]
among-individual	78.5	[59.0, 89.8]
$\log(\tau)$		
phylogenetic	42.2	[18.6, 61.2]
population	0.9	[0.0, 10.6]
among-individual	56.3	[38.0, 74.2]

Table S5: Raw data file in CSV format containing estimates of stomatal anatomy and kinetic variables associated with each curve. These are raw data for fitting multiresponse models. A final version will be deposited on Dryad after acceptance. This version is available for reviewers.

Table file:

Download from:

https://github.com/cdmuir/solanum-kinetics/raw/refs/heads/main/tables/tbl-estimates-curve.csv

Table S6: Raw data file in CSV format containing estimates of stomatal anatomy and kinetic variables associated with each population based on multiresponse model predictions. A final version will be deposited on Dryad after acceptance. This version is available for reviewers.

Table file:

Download from:

https://github.com/cdmuir/solanum-kinetics/raw/refs/heads/main/tables/tbl-estimates-accession.csv

Table S7: Fixed effect parameter estimates and 95% confidence intervals (CIs) from the posterior distribution of the selected model. For each response variable, we estimated effects of growth light intensity (sun vs. shade), measurement light intensity (high vs. low), and leaf type (amphi vs. pseudohypo). The selected model potentially includes effects of l_{gc} and f_{gmax} on τ and λ.

Parameter	**Estimate [95% CI]**
$\log(l_{\text{gc}})$	
intercept (shade, low light, amphi)	2.96 [2.79, 3.12]
effect of sun growth treatment on $\log(l_{\text{gc}})$	0.11 [0.11, 0.12]
effect of pseudohypo leaf type on $\log(l_{\text{gc}})$	-0.06 [-0.07, -0.05]
$\text{logit}(f_{\text{gmax}})$	
intercept (shade, low light, amphi)	-2.50 [-2.79, -2.20]
effect of sun growth treatment on $\text{logit}(f_{\text{gmax}})$	-0.28 [-0.35, -0.21]
effect of high measurement light intensity on $\text{logit}(f_{\text{gmax}})$	0.85 [0.82, 0.87]
effect of pseudohypo leaf type on $\text{logit}(f_{\text{gmax}})$	0.21 [0.18, 0.24]
$\log(\lambda)$	
intercept (shade, low light, amphi)	0.39 [0.29, 0.48]
effect of sun growth treatment on $\log(\lambda)$	0.11 [0.10, 0.13]
effect of high measurement light intensity on $\log(\lambda)$	0.01 [-0.01, 0.03]
effect of pseudohypo leaf type on $\log(\lambda)$	-0.04 [-0.05, -0.03]
effect of $\text{logit}(f_{\text{gmax}})$ on $\log(\lambda)$	0.10 [0.08, 0.13]
$\log(\tau)$	
intercept (shade, low light, amphi)	5.49 [5.13, 5.83]
effect of sun growth treatment on $\log(\tau)$	0.05 [0.01, 0.10]
effect of high measurement light intensity on $\log(\tau)$	0.16 [0.11, 0.22]
effect of pseudohypo leaf type on $\log(\tau)$	-0.03 [-0.05, -0.00]
effect of $\text{logit}(f_{\text{gmax}})$ on $\log(\tau)$	0.19 [0.13, 0.25]

Table S8: Random effect parameter estimates and 95% confidence intervals (CIs) from the posterior distribution of the selected model. For each response variable, we estimated among-individual random effects from repeated measures of the same individual, where possible, random effects of population (nonphylogenetic), and phylogenetically structured random effects of population. We report the random effect standard deviations (SD). For population-level random effects, we also estimated correlations between traits.

Parameter	**Estimate [95% CI]**
among-individual random effects	
SD in $\log(\lambda)$	0.08 [0.07, 0.08]
SD in $\log(\tau)$	0.21 [0.19, 0.23]
SD in $\operatorname{logit}(f_{\mathrm{gmax}})$	0.38 [0.36, 0.41]
phylogenetic random effects	
SD in $\log(\lambda)$	0.07 [0.02, 0.11]
SD in $\log(\tau)$	0.27 [0.16, 0.39]
SD in $\log(l_{\mathrm{gc}})$	0.13 [0.06, 0.22]
SD in $\operatorname{logit}(f_{\mathrm{gmax}})$	0.23 [0.07, 0.41]
correlation between in $\log(\lambda)$ and $\log(\tau)$	0.45 [-0.19, 0.83]
correlation between in $\log(\lambda)$ and $\log(l_{\mathrm{gc}})$	0.11 [-0.51, 0.68]
correlation between in $\log(\lambda)$ and $\operatorname{logit}(f_{\mathrm{gmax}})$	-0.02 [-0.65, 0.63]
correlation between in $\log(\tau)$ and $\log(l_{\mathrm{gc}})$	0.67 [0.14, 0.91]
correlation between in $\log(\tau)$ and $\operatorname{logit}(f_{\mathrm{gmax}})$	0.21 [-0.44, 0.72]
correlation between in $\log(l_{\mathrm{gc}})$ and $\operatorname{logit}(f_{\mathrm{gmax}})$	0.46 [-0.24, 0.88]
population (nonphylogenetic) random effects	
SD in $\log(\lambda)$	0.02 [0.00, 0.05]
SD in $\log(\tau)$	0.04 [0.00, 0.13]
SD in $\log(l_{\mathrm{gc}})$	0.04 [0.01, 0.08]
SD in $\operatorname{logit}(f_{\mathrm{gmax}})$	0.15 [0.07, 0.25]
correlation between in $\log(\lambda)$ and $\log(\tau)$	0.09 [-0.77, 0.83]
correlation between in $\log(\lambda)$ and $\log(l_{\mathrm{gc}})$	0.06 [-0.70, 0.74]
correlation between in $\log(\lambda)$ and $\operatorname{logit}(f_{\mathrm{gmax}})$	-0.24 [-0.84, 0.54]
correlation between in $\log(\tau)$ and $\log(l_{\mathrm{gc}})$	0.29 [-0.70, 0.89]
correlation between in $\log(\tau)$ and $\operatorname{logit}(f_{\mathrm{gmax}})$	0.11 [-0.71, 0.80]
correlation between in $\log(l_{\mathrm{gc}})$ and $\operatorname{logit}(f_{\mathrm{gmax}})$	0.39 [-0.25, 0.84]

Supporting figures

Figure file:

Download from: https://github.com/cdmuir/solanum-kinetics/raw/refs/heads/main/figures/rh-curves.pdf

Figure S1: Humidity-response curves and fitted lines. The title provides the Tomato Genetics Resource Center accession number, replicate letter, and species names. The subtitle indicates the growth light intensity (sun or shade), measurement light intensity (low or high), and leaf type (amphi or pseudohypo). Points are raw data and lines are fitted curves. The Bayesian correlation coefficient, Bayes R^2, is shown to the right of each curve.

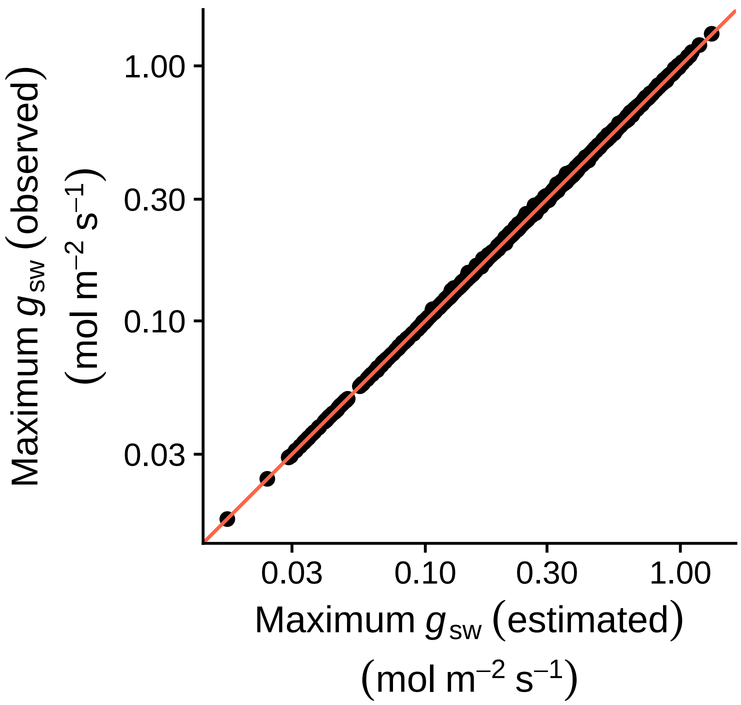

Figure S2: **The estimated (x-axis) and observed (y-axis) maximum stomatal conductances (g_{sw}) are nearly identical.** Each point is the empirical maximum g_{sw} plotted against the estimated g_{sw} for every humidity response curve. The orange line is the 1:1 line for reference. Both axes are on a log-scale.

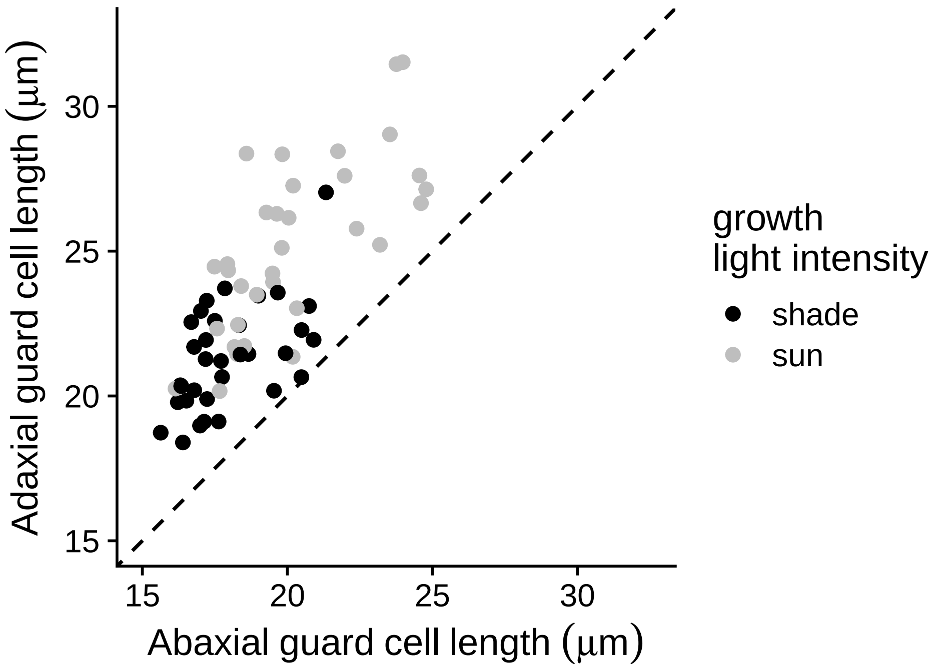

Figure S3: **Adaxial (upper) stomata are consistently larger than abaxial (lower) stomata in both sun-grown (grey points) and shade-grown (black points) plants.** Each point was calculated by taking the median guard cell length within each biological replicate and then calculating the mean among replicates within each population and growth light intensity treatment. The dashed line is the 1:1 line for reference. All points above this line indicate larger values on the adaxial surface.

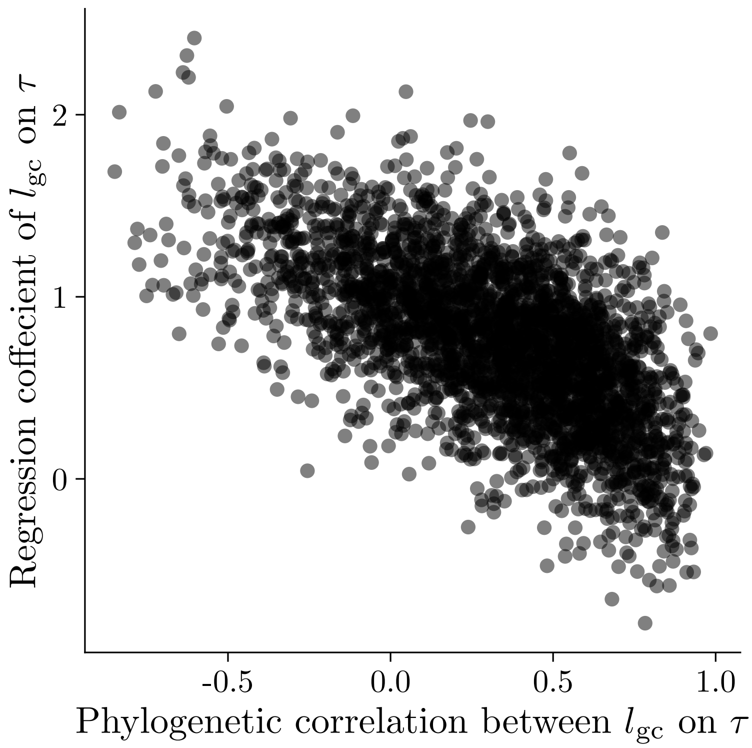

Figure S4: **Posterior estimates of phylogenetic correlations (x-axis) and fixed effects (y-axis) of guard cell length (l_{gc}) on the time-constant (τ) are colinear.** Each point is the estimate from one draw of the posterior distribution for both parameters from the model with the lowest LOOIC. The negative relationship indicates that there is tradeoff in fitting the relationship between variables as a higher level phylogenetic effect versus a lower level, individual effect. This results in more uncertainty and larger confidence intervals in each individual parameter.

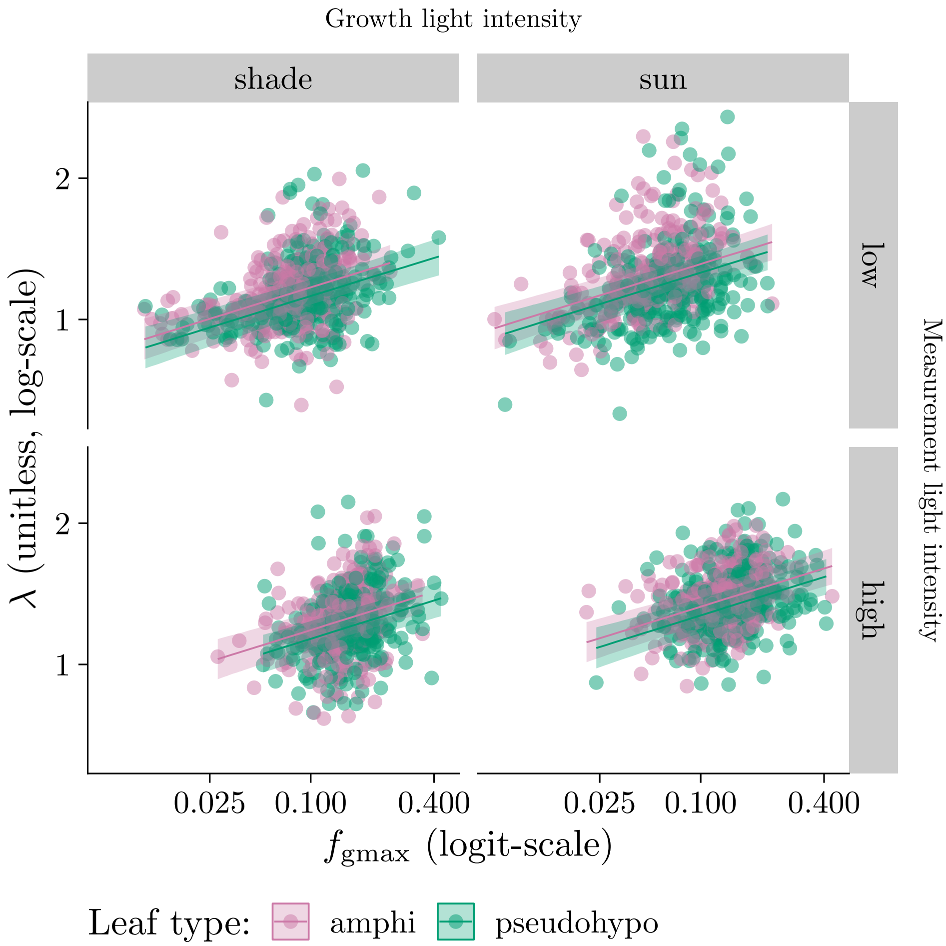

Figure S5: **Individual-level variation in stomatal apertures influences stomatal closures kinetics.** As the stomatal conductance as a fraction of anatomical maximum stomatal conductance (f_{gmax}, x-axis) increases, the stomatal closure lag-time (λ, y-axis) increases. The overall pattern was consistent across growth light intensity treatments (left and right facets), measurement light intensity treatments (top and bottom facets), and leaf types (point colors). Lines are estimated using linear regression along with 95% confidence ribbons. The growth and measurement light intensity treatments are described in the Materials and Methods section.